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Special Section: Arguing Digital History STEPHEN ROBERTSON AND LINCOLN MULLEN Arguing with Digital History: Patterns of Historical Interpretation Abstract Digital history has only rarely created interpretative or argumentative scholarship that contributes back to disciplinary understandings of the past. The reasons for this are varied: digital historians have often preferred to create scholarship for public audiences, or they have pursued forms of scholarship which do not lend themselves to explicit interpretation.

But we contend that digital historians have not had sufficient models of patterns of historical argumentation compatible with digital historical research. In this introduction, we read the articles by Rachel Midura and Leonardo Barleta that appear in this special section, and a series of other articles, in order to show the patterns of historical argumentation that digital historians can pursue. A companion website features versions of the articles by Midura and Barleta, as well as eight additional previously published articles, annotated by their authors to highlight how they developed their historical arguments so they can serve as models for argumentative digital history.

Digital history has only rarely created interpretative or argumentative scholarship in the ways that currently define both the forms and the ends of disciplinary practices of research. In terms of form, digital historians have only rarely published the interpretative (rather than methodological) journal articles or monographs that are the main mediums for research in the discipline. Perhaps this is not surprising, given that the possibilities of digital media have spurred digital historians to creativity in coming up with new forms of scholarship.

But form leads to substance. Whatever their shortcomings, the journal article and the monograph excel at disciplining historians' research so that it yields argumentative scholarship that advances new interpretations of a particular field of study. That is the end to which most conventional historical research points.

Digital history's creativity with form has not been matched by a corresponding impact on how historians understand the past.1 At times some digital historians have wondered whether their methodologically defined field ought to be making more of an interpretative impact. More often, they have celebrated—or at least justified—their freedom from being constrained to pursue the same end as other historians. Certainly, digital history has spurred a great deal of methodological

特辑：论证数字史。Stephen Robertson 和 Lincoln Mullen 的《用数字史论证：历史解释的模式》指出，数字史很少创造那种能够反哺学科内部历史理解的解释性或论证性学术成果。原因很多：数字历史学家常常更愿意为公共受众制作学术项目，或从事不容易形成明确解释的研究形式。作者认为，数字历史学家缺少与数字历史研究相兼容的历史论证模式。本文作为导言，阅读本特辑中 Rachel Midura 和 Leonardo Barleta 的文章，以及其他若干文章，展示数字历史学家可以采用的历史论证模式。配套网站收录了 Midura 和 Barleta 的文章以及另外八篇已发表论文，由作者加注说明其历史论证如何形成，以便作为论证型数字史的范例。数字史很少以当前学科研究所认可的形式和目的来生产解释性或论证性学术成果。就形式而言，数字历史学家很少发表解释性而非方法论性的期刊论文或专著；这并不奇怪，因为数字媒介的可能性推动他们创造新的学术形式。但形式会影响实质。期刊论文和专著尽管有局限，却擅长约束历史学家的研究，使之形成推进某一研究领域新解释的论证性学术。多数传统历史研究正是朝这一目标前进。数字史在形式上的创造力并没有相应转化为对历史理解的影响。一些数字历史学家曾思考，以方法界定的领域是否应产生更大的解释性影响；更多时候，他们则庆祝或至少正当化自己不必追求与其他历史学家相同目标的自由。数字史确实带来了大量方法论讨论，并把历史知识带给公共受众、教师、学生和家庭。

discussion and has brought historical knowledge to public audiences and to K–12 teachers, students, and families. Each of these is an end worth pursuing. But it is our view that digital history should—and can—make more of an impact on historical knowledge within specific fields. Historical interpretations and arguments are not the only ends that digital history should pursue, but they are an end that digital history should pursue. For several years we have sought to encourage and enable other digital historians to pursue such argumentative and interpretative scholarship. The outcomes have included an Andrew W. Mellon Foundation–funded workshop that produced a white paper on historical argumentation in digital history as well as a journal that publishes discipline-specific arguments and interpretations that come out of digital history projects in progress. This special section, the product of another Mellon-funded workshop that brought together early-career historians and the editors of this journal, represents another piece of this ongoing project. The section is accompanied by a website hosted by the Roy Rosenzweig Center for History and New Media in which authors of noteworthy journal articles, including the two articles that appear here, annotate the process by which they developed their historical arguments in order to explain how they can be models for argumentative digital history.

At the same time, other historians have been slowly amassing a body of interpretative work based on digital history. In this introduction, we offer a brief overview of the reasons that digital historians have not in the main pursued interpretative scholarship. Then we turn to an examination of several cases in which digital historians have made such arguments, including in this special section, and seek to show the patterns

of
argumentation that those articles have developed.
Our argument is straightforward. Conventional historians have many models of scholarship
in forms that pursue the end of historical argumentation.
Digital historians have many models of scholarship in forms that encourage other ends,
but they have few models of argument-driven digital history.
Therefore it is not obvious to scholars in the field, especially early-career researchers, that they can pursue historical interpretation as a primary end or how they
would go about creating such arguments. In this introduction, we seek to show how other
digital historians have pursued digital scholarship that is methodologically innovative
but also advances historical interpretations. By showing the patterns of argumentation
latent in that scholarship, we seek to encourage other digital historians to emulate
those patterns and create more interpretative scholarship.
Why Have Digital Historians Not Made Arguments?
Why digital history has not had the impact on field-specific arguments in history—an
impact on the historiography—remains something of a puzzle.
Certainly, there has been no shortage of claims by digital historians that they would
have such an impact.
However, a large proportion of digital historians have instead concentrated on speaking
directly to or with public audiences. In part that reflects the orientation of many
early digital historians toward public history and the influence of the wealth of
resulting models for such scholarship.2 Yet even beyond public history, digital
historians have made their work accessible to nonacademic

中文译文 - 第 2 页

作者认为，数字史应当并且能够对具体领域中的历史知识产生更大影响。历史解释和论证不是数字史唯一应追求的目标，但它确实应成为目标之一。多年来，作者一直试图鼓励并帮助其他数字历史学家从事这种论证性和解释性学术工作。相关成果包括 Andrew W.
Mellon 基金资助的工作坊，该工作坊产出有关数字史中历史论证的白皮书，也推动期刊发表来自数字史项目的学科性论证和解释。本特辑来自另一次 Mellon 资助工作坊，聚集早期职业历史学家和本刊编辑，是这一持续项目的又一组成部分。特辑还配有由 Roy
Rosenzweig Center for History and New
Media 托管的网站，重要论文作者会在那里注释自己形成历史论证的过程，以说明这些文章如何成为论证型数字史的模型。同时，其他历史学家也在逐渐积累以数字史为基础的解释性工作。本文先概述数字历史学家为何总体上较少追求解释性学术，再考察若干已经提出这类论证的案例，包括本特辑文章，并说明这些文章发展出的论证模式。作者的核心观点很直接：传统历史学家有许多追求历史论证目标的学术形式模型；数字历史学家也有许多鼓励其他目标的学术模型，但论证驱动的数字史模型很少。因此，尤其对早期职业研究者来说，他们能否把历史解释作为主要目标以及如何构造这类论证并不显然。作者希望通过展示其他数字历史学家如何把方法创新与历史解释结合起来，鼓励更多人仿效这些潜在论证模式。随后，文章提出问题：为什么数字历史学家没有提出更多论证？数字史为何没有对特定领域的史学论争产生更大影响，仍然是一个谜。许多数字历史学家声称会产生这种影响，但大量研究者转而面向公共受众发声，这部分源于早期数字历史与公共史的关系及其丰富范例。

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audiences to a greater extent than have their non-digital colleagues. The products of

digital research and analysis can be made available online to a far greater extent than is the case with analog history. Historical sources that have been collected and digitized, data created from sources, and exploratory visualizations and maps can all be shared with the public without making scholarly arguments and without the constraints of scholarly publication. (We ourselves have taken this approach with digital research, most notably in Digital Harlem, in the case of Stephen Robertson, and in America's Public Bible, in the case of Lincoln Mullen.) Many digital historians find audiences which are far larger than the audience of academic historians in a particular field, and they also feel that making historical resources available for open use is more rewarding and more valuable than the contribution of another journal article or monograph. In the past, such activity was seen as a kind of scholarship; with his typical prescience, Roy Rosenzweig recognized almost twenty years ago that the digital era would unsettle the relationship between historians and archival work and force us to ask, "Should the work of collecting, organizing, editing and preserving of primary sources receive the same kind of recognition and respect that it did in the earlier days of the profession?"3 The re-emergent digital forms of that practice among historians have slowly gained institutional and professional recognition, even if conventional historians too often accord it a standing that is short of the status of scholarship.

Creating digital historical resources is also labor intensive, especially at scale, so digital historians who undertake it are left little time to build such an argument. Moreover, often the exploratory analyses and visualizations that digital historians make public were never intended to be developed into argument-driven scholarship. Instead, the purpose was to find out what digital tools could do, to experiment. Tom Scheinfeldt succinctly articulated that approach in a blog post in 2010, arguing that only after "tool building, experimentation, and description" would digital tools be "sufficiently articulated and phenomena sufficiently described" for arguments to be made.4 These experiments typically focused on working with data by using a digital tool to identify features and patterns in those sources, rather than answering questions or organizing the results into an overarching argument. The work of Ben Schmidt, for instance, affords many fascinating examples of visualizations intended to explore historical sources and illustrate the potential for data analysis as a method. His map of ship paths, for example, was created "to illustrate how rich metadata alone can be as a source for historical research."5 In cases where public digital historical resources have b

ecome
the bases of scholarly arguments, they have had an indirect impact as the basis for
argumentative history without being the interpretation itself. One need look no further
than one of the earliest and most influential digital history projects: The Valley of
the Shadow.
6 The sources the site contains form the basis of Edward Ayers’s award-winning books In
the Presence of Mine Enemies and The Thin Light of Freedom.
7 While a page in the front matter of the first volume and a mention in the preface of
the second volume identifies the books as a part of The Valley of Shadow project and
points readers to the site, the books themselves offer a narrative history that shows no
indication of its digital roots and no explanation of how its argument is based on
digital methods. To take another example, Jane Kamensky’s The Exchange Artist is a
contribution to the history of the early

中文译文 - 第 3 页

数字历史学家比非数字同行更常使自己的作品面向非学术受众。数字研究和分析的成果可以比模拟时代的历史研究更容易在线公开：收集和数字化的史料、由史料生成的数据、探索性可视化和地图，都能在不形成学术论证、不受传统发表限制的情况下分享给公众。作者自己也曾采用这种路径，如 Stephen Robertson 的 Digital Harlem 和 Lincoln Mullen 的 America’s Public Bible。许多数字历史学家拥有远大于某一学术领域读者的受众，也认为开放历史资源比再发表一篇期刊论文或专著更有价值。Roy Rosenzweig 早已预见数字时代会动摇历史学家与档案工作的关系，迫使人们追问收集、组织、编辑和保存原始资料的工作是否应获得与早期职业阶段相同的认可和尊重。这种实践在数字形态中的重新出现逐渐获得制度和专业认可，即使传统历史学家往往仍不把它完全视作学术。制作数字历史资源又非常耗费劳动，尤其在大规模情况下，投入其中的数字历史学家很难再有时间构造论证。此外，许多公开的探索性分析和可视化本来就不打算发展为论证驱动的学术；目的只是实验，看看数字工具能做什么。Tom Scheinfeldt 2010年曾说，只有在工具建造、实验和描述之后，数字工具和现象才足够清楚，能够提出论证。这类实验通常关注用数字工具识别史料特征和模式，而非回答问题或组织成总体论证。Ben Schmidt 的可视化作品就是例子，他的船舶路线图展示了丰富元数据本身可如何成为历史研究来源。公共数字历史资源有时会成为学术论证的基础，但其影响往往是间接的。例如 The Valley of the Shadow 为 Edward Ayers 的两部获奖著作提供资料基础，书中却不以数字史自居，也不解释论证如何基于数字方法。Jane Kamensky 的 The Exchange Artist 也利用了波士顿 Exchange Coffee House 的3D重建，但主要呈现为美国早期共和国史著作。

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American republic, and most scholars have recognized the virtues of its narrative and
prose style. But The Exchange Artist also features plates of a 3D reconstruction of
Boston’s Exchange Coffee House, which in turn informs the prose and the analysis. Again,
the book does not present itself as a work of digital history, but it is one, at least
in part.8 It is also the case that argument-driven scholarship based on digital research
is not always visible as digital history when the emphasis is on interpretation rather

than on method. “Digital history” and especially “Digital Humanities” define themselves in an institutional sense by drawing attention to the digital against the conventional. Scholarship in Digital Humanities is noticed when there is something about its approach or method that appeals across the typical boundaries of a field or discipline. A kind of map, or maybe an approach to text analysis, might draw the attention of a twentieth-century U.S. historian, an Ottomanist, and a literary scholar for its contribution to method. But if the work is pitched primarily as a contribution to Ottoman history, then the U.S. historian is unlikely to pay it much attention, and vice versa. Put differently, what draws the interest of attendees at Digital Humanities Summer Institute or the Digital Humanities conference is very different than what draws the attention of attendees at the Organization of American Historians or the Council on Latin American History.

One thing, however, is clear. The problem with digital history’s impact on specific historical fields is no longer that editors in those fields are unwilling to publish such research. In recent years, leading journals such as the American Historical Review, the Journal of American History, the Journal of Social History, Law and History Review and others have shown that they are willing and even eager to publish scholarship in these fields, although not always in the form, or at least with the discussion of method, digital historians favor. If anything, digital history is at a premium for these journals, so long as it makes an argument recognizable as a contribution to disciplinary scholarship.

It is our contention that part of the reason this scholarship has not developed is because digital historians lack models for the kinds of work they can, and likely should, be doing. To that end, we have presented some models for how digital historians can connect their unique methodologies to historical interpretation.

Workshopping Arguments A white paper produced by participants in a workshop we convened, “Digital History and Argument,” provides a guide to the argumentative structures at work in digital collections, digital public history, digital methodological work, computational digital history, and visualizations.9 The twenty-seven authors of the white paper concluded that most forms of digital history have advanced only implicit historical arguments.

Digital collections, on the one hand, are primarily about gathering and publishing primary sources. Scholars make an implicit argument by selecting, organizing, categorizing, and describing sources, but they do not explicitly articulate those

arguments or address the implications of their collections for how the topic is understood by other scholars. The discussion of digital methodologies, on the other hand, explains and develops new ways to analyze historical sources,

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以数字研究为基础的论证性学术，在重心放在解释而非方法时，常常不被看作数字史。数字史，尤其数字人文学，在制度意义上往往通过突出数字方法与传统方法的差异来定义自身。数字人文学中的学术之所以引人注意，常因为其方法能够跨越领域边界吸引研究者；一种地图或文本分析方法可能同时吸引美国20世纪史学者、奥斯曼史学者和文学研究者。但如果作品主要定位为奥斯曼史贡献，美国史学者可能不会关注，反之亦然。换言之，数字人文学暑期学院或会议吸引人的东西，与美国历史学家组织或拉丁美洲史委员会会议所关注的并不相同。然而有一点很清楚：数字史对具体历史领域影响不足，问题已不再是领域期刊不愿发表此类研究。近年 American Historical Review、Journal of American History、Journal of Social History、Law and History Review 等重要期刊都愿意甚至渴望发表相关学术，只要它提出能被识别为学科贡献的论证。作者认为，问题的一部分在于数字历史学家缺少可供参照的工作模型。为此，作者提出若干模型，说明数字历史学家如何把独特方法与历史解释连接起来。文章随后讨论“工作坊中的论证”。作者召集的工作坊所产白皮书《数字史与论证》为数字收藏、数字公共史、数字方法论工作、计算数字史和可视化中的论证结构提供指南。白皮书27位作者认为，多数数字史形式只提出了隐含的历史论证。数字收藏主要收集和发布原始资料；学者通过选择、组织、分类和描述史料提出隐含论证，但通常不明说这些论证，也不讨论馆藏如何改变其他学者对主题的理解。数字方法论讨论则解释和发展新的史料分析方式，但方法本身并不是解释。

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but again, those methods are not the interpretations themselves. Because digital public history is created both for public audiences and often in collaboration with those communities, it primarily engages in a conversation with the public rather than with other scholars, and it prefers implicit interpretation rather than explicit argumentation.
Computational digital history and digital history visualizations are more amenable to explicit argument, as they both create a broad view of a set of sources that highlights patterns that can be interpreted. Word frequency analysis, topic modeling, and word-embedded modeling identify quantitative patterns in textual data, while centrality and betweenness calculations identify quantitative patterns in networks. Maps and network visualizations identify spatial patterns in data. But projects involving both forms of digital history often present those quantitative analyses and visualizations with little interpretation. By contrast, the argumentative interpretation that appears in the pages of scholarly historical journals such as the Journal of Social History makes explicit its claims, evidence, logic, and engagement with other scholarship. Examining why there are so few examples of digital history that make such explicit arguments, the authors of the white paper concluded that scholars lack conceptual models of how to apply digital methods to historical questions.

To encourage more argument-driven digital history we collaborated with the Andrew W.
Mellon Foundation to organize a series of workshops to support authors in the process of
writing, peer review, and publication. The two articles that appear in this special
issue/section are the result of that process. To extend the utility of these articles,
we also created a site, Models of Argument-Driven Digital History, hosting a freely
available version of the articles annotated to serve as models of how to conceive and
construct interpretations and arguments using digital history methods and materials.
When the changed circumstances of the pandemic curtailed plans to discuss the process
and models in a session at the American Historical Association annual meeting, we moved
instead to expand the site with eight additional articles, published scholarship that
provides successful models of argument-driven digital history.10 Patterns of Argument-
Driven Digital History In this section we analyze a set of articles which have made
field-specific arguments on the basis of digital historical work. In doing so, we seek
to show two things. First, it is indeed possible for digital historians to make
significant arguments in their historical fields. However, as we previously mentioned,
such work often does not get marked as digital history because its contributions are
interpretative rather than methodological, and because it is published in journals for
specific fields rather than in Digital Humanities journals. So we have attempted to read
broadly across fields other than our own to identify examples of digital historical
arguments, though of course we have undoubtedly missed many worthy examples in other
fields or languages other than English. Second, we have tried to read these articles to
show how they can serve as models for other historians to make similar arguments.
Advances in the historical discipline often occur when specific essays provide models
for how other historians can offer new interpretations: one thinks of Robert Darnton’s
“The Great Cat Massacre” or Joan Scott’s “Gender: A Useful Category of Analysis.”11
While

数字公共史面向公众并常与社群合作，因此主要与公众而非其他学者对话，也倾向于隐含解释而非明确论证。计算数字史和数字史可视化更容易形成明确论证，因为它们为一组史料创造广阔视图，突出可解释的模式。词频分析、主题模型和词嵌入模型可识别文本数据中的量化模式，中心性和中介性计算可识别网络中的量化模式，地图和网络可视化则识别空间模式。但许多项目呈现量化分析和可视化时解释不足。相比之下，Journal of Social History 等学术期刊中的论证性解释会明确提出主张、证据、逻辑和与既有研究的关系。白皮书作者认为，数字史明确论证少，是因为学者缺少把数字方法用于历史问题的概念模型。为鼓励更多论证驱动的数字史，作者与 Mellon 基金合作组织工作坊，支持作者写作、同行评议和发表。本特辑两篇文章就是该过程产物。为扩大这些文章的作用，作者还创建

Models of Argument-Driven Digital History
网站，提供带注释的开放版本，作为使用数字史方法和材料构思并构建解释和论证的范例。疫情改变了原定会议计划后，他们又扩展网站，加入八篇已发表的成功论证驱动数字史文章。接下来，文章讨论论证驱动数字史的模式。作者分析一组基于数字历史工作提出具体领域论证的文章，旨在说明两点：第一，数字历史学家确实可以在自己的历史领域提出重要论证；第二，这些文章可作为他人提出类似论证的模型。历史学进展常来自具体论文提供新的解释模型，如 Darnton 的《猫大屠杀》和 Joan Scott 的《性别：一个有用的分析范畴》。

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perhaps none of the articles discussed here rises to the level of those celebrated works of scholarship, they are useful models for digital historians seeking to make arguments in different fields on the basis of similar kinds of sources and methods.

Published digital history that successfully makes explicit arguments shares several broad structures, with variations associated with different digital methods. The selection of evidence with which the process of historical argument begins is framed more broadly in digital history, including fragmentary sources not amenable to close reading and sets of sources expanded in scale and comprehensiveness. That foundation supports arguments centrally concerned with context, and that develop by moving across scales, often beginning with the broadest picture. (Here digital history differs from conventional historical arguments, which tend to move from the particular to its context.) Tim Hitchcock has described that as a "data-first" approach, in which the historian "purposefully presents a comprehensive set of data, on the basis of which the reader is guided to a specific conclusion" as opposed to the typical approach of the conventional historian, which centers "a claim or argument, evidenced through a narrow selection of precisely relevant data."12 Digital methods provide an empirically grounded picture of the typical and representative in contrast to outliers, allowing for a systematic visual or quantitative examination of patterns through correlation, resemblance, or proximity. The sources selected through this process as the most important are the subject of close reading to address interpretive questions that the digital methods being used cannot answer, reengaging with the complexities of sources necessarily simplified to create data.

The forms of digital history most amenable to explicit disciplinary arguments are visualizations of data such as maps, networks and 3D models, and computational analysis of data. Far from mutually exclusive, those forms are often used in combination to produce network visualizations analyzed using algorithms and computational text analysis presented as maps and network graphs.

Digital historians developing arguments based on spatial visualizations are building on historians' long-standing use of mapping as a conceptual framework, albeit from

a
metaphorical perspective: the use of “the idiom of borders and boundaries, frontiers and
crossroads, centers and margins,” as Karen Halttunen put it.13 Moreover, maps conveying
information are a longstanding feature of historical publications, although generally
they are limited to identifying locations. Digital history spatial visualizations by
contrast show data in a spatial context, expanding the scale of information being
conveyed, incorporating fragmentary sources, combining layers of different kinds of
sources, and facilitating analysis at multiple scales. In employing qualitative sources
and not relying on statistical analyses, these visualizations differ from mapping using
Historical GIS. In “Seeing Emancipation: Scale and Freedom in the American South,”
Edward Ayers and Scott Nesbit mapped locations and movements of Union troops compiled by
Frederick Dyer, and combined them with locations of what they defined as “emancipation
events” gleaned from the Official Records of the War of the Rebellion, together with
several newspapers and diaries. This map put together the pieces of the “vast, distended
and chaotic” process of emancipation that unfolded within shifting boundaries and at an
uneven pace to provide a picture that showed “the patterns, proportions and timing of
emancipation,” and “how actions overlap, penetrate, and conflict with each other.”14
Ayers and

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虽然本文讨论的文章未必达到这些经典作品的高度，但它们对希望基于类似史料和方法在不同领域提出论证的数字历史学家很有用。成功提出明确论证的数字史有几个共同结构，并随方法不同而变化。历史论证的证据选择在数字史中通常被更宽泛地框定，包括不适合细读的碎片化史料，以及在规模和完整性上被扩展的史料集合。这一基础支持以语境为中心的论证，并通常从最广阔的图景开始，在不同尺度之间移动。Tim Hitchcock 称之为“数据优先”路径：
历史学家有意识地呈现一组综合数据，并引导读者从中走向特定结论，而不是像传统历史学家那样从一个主张出发，只选取精确相关的狭窄证据。数字方法提供了关于典型、代表性和异常值的经验基础图景，使研究者能够通过相关性、相似性或邻近性系统考察视觉或量化模式。经此过程筛选出的重要史料仍需细读，以回答数字方法无法解决的解释性问题，并重新面对那些为生成数据而被简化的史料复杂性。最适合明确学科论证的数字史形式包括地图、网络、3D 模型等数据可视化，以及数据的计算分析。这些形式并不相互排斥，常常结合使用，例如算法分析的网络可视化，或以地图和网络图呈现的计算文本分析。基于空间可视化提出论证的数字历史学家延续了历史学长期把地图作为概念框架的传统，即边界、边疆、十字路口、中心和边缘等语汇。传统历史出版物也长期使用地图，但多限于标示地点；数字史空间可视化则把数据置于空间语境中，扩大信息尺度，吸收碎片化史料，组合多类史料层，并促进多尺度分析。

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Nesbit progressively shift the scale of their analysis to examine specific patterns
within the map, selecting first the entire Civil War South in summer 1864 to show how
“military movements overlay legal and demographic geographies to create a complex

terrain for emancipation," then narrowing the focus to Virginia to highlight "a loose coordination between enslaved men and women fleeing slavery and the paths of large armies," switching to a narrative that shows that "armies were unreliable vehicles for emancipation, bringing heartbreak as well as liberation."15 In this argument digital mapping provides an empirically grounded, more comprehensive picture of emancipation and its relationship to the presence of Union troops, making visible patterns in the proximity of the two types of events that are then explored at different scales to show the impact of varying contexts and the deeply contingent nature of emancipation.
A further form of argument based on digital mapping uses quantification to identify patterns beyond those made visible on the map.16 Harmony Bench and Kate Elswit, in their "Katherine Dunham's Global Method and the Embodied Politics of Dance's Everyday," quantify movement and examine it at multiple scales. By using performance contracts, receipt books, personal logs and activity diaries, programs, newspaper clippings, and personal and professional correspondence to reconstruct where Dunham was every day between January 1, 1950 and December 31, 1953, they generated an analysis "counter to dominant historical approaches that take a midfield view to build broad narratives through a small set of exemplary moments, anecdotally illustrating an argument."17 Their mapping of travel between locations and timeline of the duration of stays reveal patterns in touring that "rebalance the geography of Dunham scholarship," revealing the importance of cities beyond those featured in existing scholarship and of nightclubs as performance venues, new contexts in which to interpret her choreographic and ethnographic work. Bench and Elswit then turn to close reading to highlight the embodied labor involved in the movement, which a visualization of points connected by lines tends to obscure. They use the paradigm of friction to elaborate the difficulties Dunham encountered moving, including available modes of transportation, the logistics of crossing borders, racial restrictions on accommodations, and the constant need to exchange currencies. Those pressures wore not only on Dunham's mobility, but on her body. Granular data that reveal Dunham performing or rehearsing 74 percent of the days in the period under study, and traveling for more than half of the remaining time, directs attention to the wear and tear on Dunham's body and the fatigue and physical complaints that she wrote about. As in Ayers and Nesbit's argument, mapping here provides an empirically grounded, more comprehensive picture, with the additional use of quantification of the number and duration of visits to locations to identify previously unrecognized patterns that are then analyzed. In also exploring those patterns a

t
different scales, Bench and Elswit go closer than Ayers and Nesbit to examine th
e human
scale, an individual's experience of the movement visualized on the map.
The 3D reconstruction also provides the basis for arguments about movement at th
e human
scale. The historian using that method locates individuals within a digital mode
l of a
space and then analyzes archival evidence of their activity in that context. In
"Reconsidering Poor Law Institutions by Virtually Reconstructing and Re-Viewing
an
Eighteenth-Century Workhouse," Susannah Ottaway and Austin Mason created a model
 of the
House of Industry at

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Ayers 和 Nesbit 的《看见解放》把 Dyer 编纂的联邦军队位置和移动，
与《叛乱战争官方记录》、报纸和日记中提取的"解放事件"位置结合起来。地图把解放这一巨
大、分散且混乱的过程拼合起来，显示其模式、比例、时机以及不同行动如何重叠、渗透和冲突
。他们不断改变分析尺度，先看1864年夏整个内战南方，说明军事移动如何叠加法律和人口
地理，构成复杂的解放地形；再缩小到弗吉尼亚，展示逃离奴隶制的奴隶与大军路径之间的松散
协调，同时叙述军队作为解放载体并不可靠，既带来自由也带来心碎。这里，数字制图提供了经
验基础更坚实、更全面的解放图景，使联邦军出现与解放事件邻近性的模式可见，并通过不同尺
度揭示语境差异和解放深刻的偶然性。另一种基于数字地图的论证使用量化来识别地图上可见模
式之外的模式。Harmony Bench 和 Kate Elswit 在研究
Katherine Dunham
的全球方法和舞蹈日常政治时，利用合同、收据、日志、日程、节目单、剪报和通信重建
Dunham 在1950至1953年每一天的位置，量化移动并在多尺度上考察。旅行地图
和停留时间线揭示了巡演模式，重新平衡 Dunham 研究的地理，显示既有研究忽视的城
市和夜总会场所的重要性。随后他们通过细读呈现地图点线可视化容易遮蔽的身体劳动：交通方
式、跨境后勤、住宿中的种族限制和频繁换汇都对 Dunham 的身体造成压力。数据显示
她在研究期内74%的日子演出或排练，剩余时间一半以上在旅行，从而指向身体损耗、疲劳和
疼痛。

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Gressenhall, Norfolk, that allowed "human-scale exploration using a first-person
controller in the mode of commercial video games." They then deployed this model
 as a
"virtual 'stage' for [their] historical actors."18 Their argument develops by fi
rst
examining the features of the building and then tracing the routes of the poor a
nd
guardians as they made their way to workrooms, dining spaces, the schoolroom, co
mmittee
room, and attendant spaces. Ottaway and Mason show that the building reinforced
paupers'
ease of movement in and out of the workhouse and authorized the role and power o
f the
guardians articulated in documents. At the same time, the building facilitated a
 mixing
of different categories of the poor that was at odds with the authorities' commi
tment to
the segregation of groups. This combination reveals that the building embodied h
ow "new
ambitions around policing were actively restrained by the expectations and actio
ns of
the poor in the last decades of the eighteenth century."19 While creating a cont
ext is
the foundation of this argument, it is employed differently here than by Ayers a

nd
Nesbit and Bench and Elswit; additional evidence is put in the context of the mo del,
rather than patterns being identified within it. The argument develops by explor ing the
relationship between evidence of different activity and movement and the buildin g model
to show the gap between ideal and practice.20 Leo Barleta's article in this spec ial
issue, "Spatial Genealogies Mobility, Settlement, and Empire-Building in the Bra zilian
Backlands, 1650–1800," takes a different approach to patterns of movement, visua lizing
journeys as a network of relationships that highlights that kinship not geograph y or
infrastructure was key to mobility.21 His sources are genealogical compilations, containing thousands of brief biographies, fragmentary sources mined for places of birth
and marriage that could be used to trace mobility.22 Counter to arguments that e mphasize
the exceptionality of the journeys of intrepid explorers, Barleta's visualizatio ns
highlight the repetition and routinization of movement between coast and interio r, with
the thickness of the lines linking points indicating the volume of traffic (whic h Bench
and Elswit visualized as a timeline). The way that the lines fill the resulting image
visually conveys his argument that the space was connected and made cohesive thr ough
family ties, that "these colonists' sustained mobility wove together an otherwis e
fragmented territory." To elaborate the texture and social aspects of the lives captured
in these patterns, Barleta turns to snippets of an individual life story, constr ucted
from other sources. The argument thus follows the structure of establishing a co ntext of
journeys that makes visible patterns that are then analyzed.
Networks, like maps, have been widely used metaphorically by historians as a con ceptual
framework. The language of networks is applied, as Ahnert et al.
note, to "communities of practitioners, the dissemination of ideas, or the relat ionship
between certain texts, images or artifacts."23 Digital historians' network
visualizations show subjects in the context of relationships, often with a secon dary use
of computational measures to identify for closer analysis those highly connected and
those who by connecting the most individuals are important to the structure of t he
network. In "Kaskaskia Social Network: Kinship and Assimilation in the French-Il linois
Borderlands, 1695–1735," Robert Morrissey uses a network created from marriage a nd
baptismal records to understand the function over time of marriages between Fren ch men
and Native American women in Illinois country in the early eighteenth century an d "the
communities

3D 重建也可以支撑人类尺度上的移动论证。使用这种方法的历史学家把个人置于数字空间模型中，再分析其活动的档案证据。Susannah Ottaway 和 Austin Mason 重建了 Norfolk Gressenhall 的 House of Industry，使研究者能以类似电子游戏第一人称控制器的方式进行人类尺度探索，并把模型用作历史行动者的虚拟舞台。他们先考察建筑特征，再追踪穷人和管理者到工作间、餐厅、学校、委员会室和附属空间的路线，说明建筑如何既加强贫民进出济贫院的便利性，也确认档案中所见管理者权力；同时，建筑又使不同贫民类别混合，违背当局隔离人群的承诺。这揭示出18世纪末围绕治安管理的新抱负如何受到穷人期待与行动的主动限制。在这里，模型是论证基础，但不同于前述地图案例：附加证据被放入模型语境，而不是在模型内部识别模式；论证通过比较活动和移动证据与建筑模型之间关系，显示理想与实践的差距。Leonardo Barleta 本特辑文章则用另一方式处理移动模式，把旅程可视化为关系网络，突出亲属关系而非地理或基础设施才是流动关键。他的资料来自谱牒汇编，其中成千上万条简短传记提供出生地和婚姻地点。与强调无畏探险者非凡旅程的论点不同，Barleta 的可视化突出海岸与内陆之间移动的重复和常规化；线条粗细显示交通量，使读者直观看到家族纽带如何把破碎领土连成整体。他随后转向个体生命故事片段，以展现这些模式中人的生活质地和社会面向。

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and identities they sometimes created."24 Morrissey develops his argument through a series of network visualizations for two time periods, using the patterns that are made visible to identify highly connected individuals for close analysis.

In the first period patterns can be relatively easily seen in a graph of godparents and marriages, and a graph limited to godmothers, with directional links from godmother to parents. The most highly connected men and women in the network were those most oriented toward the agrarian life considered French. To more clearly see the social prominence and influence of leading families he switches to computational measures of centrality, betweenness scores (how many people are connected through a certain individual) combined with additional quantitative data about the land they had under cultivation and their household size. The results of that analysis show that the most connected and those that connected the most individuals had the greatest involvement in farming. The network's picture of kinship ties exerting pressure toward French agrarian cultures is very different from a historiography that argues that mixed race fur trade marriages were instead "rooted in an indigenous framework and an indigenous sphere."25 In this argument the network graph provides a comprehensive context of relationships, making visible patterns of connections.

Quantitative measures are then used to identify patterns within those connections in terms of who was connected to the most individuals and who connected the most individuals.26 Arguments focused on complex relationships rely more heavily on quantitative measures. The network of epistolary data that Ruth Ahnert and Sebastian Ahnert develop in their "Protestant Letter Networks in the Reign of Mary I: A Quantitative Approach" included, in addition to sender and recipient, other relationships mentioned in the correspondence: requested links, implied links, reported

links, messenger links, spousal links, and other family links.27 While the whole network
graph confirms the importance of the martyrs as hubs in the community, plotting the
taxonomy of links identifies another large group of people who are highly connected
despite sending or receiving relatively few letters.
To understand the role of such individuals in the network, Ahnert and Ahnert calculated
not just those who connected the most individuals but also how many well-connected
individuals an individual was adjacent to in the network
(a measure called eigenvector centrality). High scores in those measurements identified
men, and more surprisingly women, who provided the infrastructure of the network by
carrying letters and delivering financial support. To bring this pattern further into
focus, “to identify general rules for the overall structure and function of the
network,” Ahnert and Ahnert next developed quantitative criteria to establish categories
of roles in the network, grouping them into three categories of “leader,” three
categories of network “sustainer” (including financial supporters and letter couriers),
and one for a peripheral figure. A temporal examination of the network highlights that
it was able to endure a systematic attack that removed fourteen of the twenty most
central individuals thanks to the survival of these lesser-known figures who provided
its infrastructural backbone.
While the broad structure of this argument echoes that of Morrissey in establishing a
context of relationships, it relies more on quantitative calculations to identify
previously unrecognized patterns for analysis and to analyze the role in the network of
the individuals prominent in those patterns.

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网络像地图一样，也长期被历史学家作为概念框架使用。Ahnert 等人指出，网络语言常用于实践共同体、观念传播或文本、图像、器物之间的关系。数字历史学家的网络可视化把对象置于关系语境中，并常用计算指标识别高度连接者和对网络结构重要的桥接者。Robert Morrissey 的 Kaskaskia 社会网络利用婚姻和洗礼记录，研究18世纪初伊利诺伊地区法国男性与原住民女性通婚如何随时间作用，以及它们有时创造的共同体和身份。他通过两个时期的网络可视化发展论证，并用可见模式找出应细读的高度连接个体。在第一阶段，教父母与婚姻关系图、以及只看教母且从教母指向父母的有向图，使模式较易看出。网络中最高度连接的男女，是最趋向被视为法式农业生活的人。为更清楚地观察主要家庭的社会地位和影响，他转向中心性、中介性等计算指标，并结合耕地和家庭规模等量化数据。结果显示，最相连和最能连接他人的个体也最深参与农业。网络呈现的亲属纽带推动法式农业文化的图景，与认为混血毛皮贸易婚姻根植于原住民框架和领域的史学观点不同。这个论证中，网络图提供关系的综合语境并使连接模式可见，再用量化指标识别谁连接最多以及谁把最多人连接起来。Ruth Ahnert 和 Sebastian Ahnert 关于玛丽一世时期新教书信网络的研究，进一步依赖量化指标。除收发信人外，他们还纳入书信中提到的请求关系、隐含关系、报道关系、信使关系、配偶关系和其他家庭关系。

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Rachel Midura’s article in this special section, “Itinerating Europe: Early Modern

Spatial Networks in Printed Itineraries, 1545–1700," uses networks to analyze itineraries, a fragmentary source that cannot be effectively analyzed with close reading. Since these documents contain geographical information, mapping might seem to be the appropriate form of visualization, but itineraries "presented cities as semi-spatial, abstract concepts and connections."28 The abstract space of the network is a better fit than the geographical space of the map, preserving "the elements of hierarchy, directionality, and centrality conveyed by the original itinerary format."29 Midura has created a network that places routes in context, allowing her to visually and quantitatively identify anomalies and demonstrate long term trends.30 The relative importance of cities within the itinerary corpus is first analyzed visually by examining the number of connections to locations in the whole network. Following the structure of argument common to network graph analysis, Midura turns to centrality metrics for a more nuanced approach, calculating both betweenness and eigenvector scores.

The results are visualized on a map to guide the analysis in the direction of real journeys, as each measure highlights a different feature of the routes: their inclusion of pilgrimage routes and the structuring effect of the mountain passes. However, adding a temporal dimension, moving to a dynamic network model, reveals that both inherited pilgrimage and mercantile routes "played a diminishing structural role in the overall conceptual network."31 The life of a route was based on the period of its active publication in itineraries; analysis of route data and close reading identifies four clear periods, which are refined by creating a dynamic network model and extracting static networks to represent slices of time. Tracing the changes in the network, Midura finds the most dramatic shift near the end of the period, when "the cities included, the types of connections between them, and the overall structure of the conceived European systems of communication, travel, and exchange were altered."32 Midura's argument comes out of making a connection between the historical phenomenon she studies– travel and the relationship between cities–and the way that phenomenon can be measured, visualized, and above all modeled by the networks she creates. As with other network-based arguments, Midura creates a broad context of relationships, in this case between cities, and then uses quantitative measures to identify and analyze patterns in that visualization. As she points out in the annotated version of the article, that analysis of the graph "relies on a spectrum of interpretations and visualizations, matching the approach to the question being asked in each section."33 Digital historians developing arguments based on computational text analysis lack an existing conceptual framework of th

e kind
available for spatial visualizations and network graphs. The labels “distant rea
ding”
and “machine reading” suggest a relationship between computational text analysis
 and
“close reading,” an approach commonly used in humanities research. However, wher
e
digital maps and network graphs visualize data related to the concepts historian
s have
used metaphorically, the results of computational text analysis have no such con
nection
with close reading. The creation of data breaks text into words, with little or
no
context depending on how many consecutive words are grouped together. Since cont
ext is
central to determining the meaning of words, the patterns found in the quantitat
ive
data, whether presented in charts, maps, or networks, in the first instance are
not
related to meaning. Text analysis thus

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整体网络图确认殉道者是共同体中的枢纽，但对关系类型进行绘制又识别出另一大群人：他们虽然收发信较少，却高度连接。为理解这些人在网络中的作用，Ahnert 夫妇不仅计算谁连接最多人，还计算某人毗邻多少连接良好的个体，即特征向量中心性。高分个体包括男性，也出人意料地包括女性，他们通过携带信件和传递经济支持为网络提供基础设施。为了进一步识别网络整体结构和功能的一般规则，他们制定量化标准，把角色分成领导者、网络维持者（包括经济支持者和信件传递者）以及边缘人物。时间维度分析显示，即使二十位最核心人物中十四位被系统性攻击移除，网络仍能存续，正是因为这些不太知名的人提供了基础骨干。该论证总体上与 Morrissey 类似，都建立关系语境，但更依赖量化计算来识别此前未被注意的模式，并分析模式中突出个体在网络中的作用。Rachel Midura 本特辑文章《行旅欧洲》使用网络分析1545至1700年的印刷行程表。这些碎片化来源不能有效细读；虽然包含地理信息，似乎适合地图，但行程表把城市呈现为半空间的抽象概念和连接。因此，网络的抽象空间比地图的地理空间更适合，能保存原始行程表中的等级性、方向性和中心性。Midura 创建网络，把路线放入语境，既可视觉地也可量化地识别异常并展示长期趋势。她先通过查看整个网络中地点连接数量分析城市相对重要性，再使用中介性和特征向量中心性进行细化。结果再被绘制到地图上，引导分析真实旅程：不同指标揭示朝圣路线和山口的结构作用。加入时间维度后，动态网络模型显示传统朝圣和商业路线在整体概念网络中的结构作用逐渐减弱。

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relies more heavily than other digital methods on close reading to connect patte
rns with
meaning.34 As a result, this method is generally used to examine large collectio
ns of
texts—more material than an individual could analyze with close reading.
One result of beginning with large datasets has been arguments focused on texts
 as
objects of study, which begin with “a corpus looking for a question.” Tim Hitchc
ock and
William Turkel used word counts to reexamine the evidence the Old Bailey Proceed
ings
provides of courtroom procedure; historians, based on a combination of close rea
ding and
statistical sampling, have assumed the length of trial reports reflects legal pr
actices,
such as very quick trials.35 Hitchcock and Turkel analyzed the entire Proceeding
s as a
“single massive text object,” beginning by charting the number of words per year
 and

trials per year.
Read in relation to the existing literature, the chart generally fits conventional pictures of the eighteenth century, but is at odds with those of the nineteenth century, pointing to a series of specific moments that deserve further investigation. Shifting to an analysis of the Proceedings as a collection of trials, Hitchcock and Turkel analyze charts that show spasmodic changes in distribution of eighteenth-century trial lengths that indicate that they changed in response to factors outside court business. As such, they have limited value as evidence for the rise of the adversarial trial. By contrast, nineteenth-century trials show a gradual pattern of change and consistent pattern of reporting that suggests that they do reflect courtroom practice, and fit arguments about the rise of plea-bargaining in this period. Hitchcock and Turkel then examine specific factors associated with trials reported at different lengths: the seriousness of the offense, the verdict, and conviction rates. That quantitative analysis, possible because the digital text of the Proceedings is marked up with tags that identify information about those factors, confirms the two different regimes of trial reporting and the impact of the rise of plea bargaining. In using a comprehensive source to understand the nature of the source itself, not as a context for patterns visible within it, this form of argument focuses on identifying broad patterns across time rather than analyzing their details. It develops by applying different lenses that capture various facets of the data rather than shifting scale to close reading of material that forms patterns. As with Midura's argument, Hitchcock and Turkel were able to make a successful argument because they identified something about their dataset that could be measured (the word count of trials) which was a reasonable proxy for a historical change (the length and structure of legal proceedings), and then corroborated that change over time with other forms of historical knowledge.
A similar focus on identifying broad patterns shapes arguments that use topic modeling, the creation of groups of frequently collocated words, to identify some of the meaning of texts. Sharon Block and David Newman analyze tens of thousands of historical abstracts using a combination of word frequency and topic modeling to determine "what place does women's history have in the field at large and what kind of subjects are included within women's history; where in various regional histories does women's history most frequently appear; when do publications on women's history increase or decrease in numbers as well as when, chronologically, women's historians most fo

cus
their efforts.”36 The argument proceeds through three scales of analysis, calculating
the amount of women’s history in all the abstracts, and then using topic modeling to
identify the

在 Midura 的论证中，一条路线的生命被界定为它在行程表中活跃出版的时期；路线数据分析和细读识别出四个时期，再通过动态网络模型和静态时间切片加以细化。追踪网络变化，她发现最剧烈的转变出现在研究期末，城市选择、连接类型以及欧洲通信、旅行和交换体系的整体构想都发生了改变。Midura 的论证来自把所研究的历史现象（旅行和城市关系）与能够测量、可视化并建模这一现象的网络联系起来。与其他网络论证一样，她先建立关系的广阔语境，再用量化指标识别和分析可视化中的模式。她也指出，图形分析依赖一系列解释和可视化，需要根据每一节提出的问题匹配方法。相比之下，基于计算文本分析的论证缺少空间可视化和网络图那样现成的概念框架。“远读”和“机器阅读”等标签暗示了计算文本分析与人文学细读的关系，但数字地图和网络图可视化的是历史学家已用隐喻讨论的概念，而计算文本分析的结果与细读没有这种直接联系。把文本转成数据会把文本打碎为词，视连续词组长度而只有很少或没有语境。由于语境对词义判定至关重要，量化数据中发现的模式无论以图表、地图还是网络呈现，最初都不直接等于意义。因此，文本分析比其他数字方法更依赖细读来把模式与意义连接起来。这种方法通常用于分析个人无法细读的大型文本集合。以大数据集为起点的结果之一，是把文本本身作为研究对象的论证，即从“语料库寻找问题”开始。Tim Hitchcock 和 William Turkel 用词数重新考察 Old Bailey Proceedings 关于法庭程序的证据，历史学家此前基于细读和统计抽样，常假定审判报告长度反映快速审判等法律实践。他们把整个 Proceedings 当作一个巨大文本对象，先绘制每年词数和审判数。

subject areas in abstracts identified as women’s history, concluding with a case study
of one theme, the history of sexuality, with quantitative evidence of patterns presented
in charts and tables. The broad pattern in which “different thematic emphases
predominate in various regional specialties within women’s history” is echoed in the
case study (5 of the 40 topics, 9 percent of women’s history abstracts), in which the
amount and content related to sexuality varies more by region than chronology. Jo Guldi
makes similar use of topic modeling to establish broad patterns in the parliamentary
debates of Hansard—“new tensions and turning points that characterized the uptake of
infrastructure over the longue duree.”37 In a model of 500 topics, a level that returned
an overview of parliamentary business, Guldi looked for “words about technology,
infrastructure, and the built environment, hoping to find patterns that replicate,
enhance, or nuance the guns-maps-and-steel thesis of earlier scholars.”38 She chose a
subset of topics and analyzed them as dynamic topic models, comparing the words and
their relative ranking in the topic in different decades in tables shaded to highlight
patterns. These topic models point to “significant contexts,”
“highlighting moments of shifting technology, institutional alignments, and political
structures in the landscape.”39 Like arguments based on word frequency, these arguments

based on topic modeling use a comprehensive dataset to understand the nature of the
source itself, not as a context for patterns visible within it. They differ from
arguments like that of Hitchcock and Turkel in developing their interpretation by
shifting scale, to focus on particular topics rather than the text as a whole, and then
on change over time, reflecting how topic models initially are not broken down
temporally but need to be additionally analyzed as dynamic topic models.
A second strand of computational textual analysis approaches large datasets with
specific questions and develops arguments that extend to close reading.
Digital historians analyzing subjects that are denoted with proper nouns—people, places,
organizations, products—have constructed arguments about those subjects based on word
frequency. Cameron Blevins "measure[s] the frequency and distribution of specific
geographic place-names across the pages of the Houston Daily Post" to examine "how
newspapers construct space in an age of nationalizing forces."40 While newspapers are a
familiar source, Blevins's argument in "Space, Nation and the Triumph of Region" is
based on evidence of a size and comprehensiveness made possible by digital methods; an
examination of 1700 issues and aggregation of all the contents of those issues, from
news to classified ads. He structures the argument in terms of the two distinct scales
of the paper's imagined geography that appear when the frequencies with which places
appear are visualized on maps: "a national scale oriented toward New York and the
American Midwest, and a dominant regional scale of Texas and its immediate orbit." At
the national scale, Blevins placed that "imagined" geography in the context of "actual"
geography to highlight anomalies: cities that appear more than their population size
would predict as well as an absence of the American South. Overlaying a map of the
railroad network indicates that places that feature in the paper are oriented to that
network. Blevins shifts the scale of his analysis to confirm that finding. First, he
provides a close reading that reveals a recurring section of the newspaper devoted to
railroads. Then a sampled content analysis, mapping what types of content appeared in
what parts of each issue of the paper, demonstrates a larger relative presence of New
York and

将这些图表与既有研究联系起来看，它们大体符合18世纪的常规理解，却与19世纪图景不合，指出一些需要进一步调查的具体时刻。当把 Proceedings 作为审判集合来分析时，Hitchcock 和 Turkel 发现18世纪审判长度分布的突变显示其受法庭业务外因素影响，因此作为对抗式审判兴起证据的价值有限。相反，19世纪审判显示渐进变化和稳定报道模式，似乎反映了法庭实践，并符合关于认罪协商兴起的论证。他们随后考察不同长度审判报告所关联的因素，如罪行严重性、裁决和定罪率。由于数字文本带有标记，可识别这些因素，这种量化分析确认了两种审判报道制度和认罪协商兴起的影响。这里

的论证用综合来源理解来源本身的性质，而不是把它作为内部模式的语境；它关注跨时间识别宽泛模式，并通过不同透镜捕捉数据不同侧面，而不是转向构成模式的材料进行细读。Hitchcock 和 Turkel 的成功在于，他们找到了可测量的东西（审判词数）作为历史变化（法律程序长度和结构）的合理代理，并用其他历史知识加以佐证。类似的宽泛模式识别也塑造了主题模型论证。主题模型通过找出经常共现的词群来识别文本意义的一部分。Sharon Block 和 David Newman 用词频和主题模型分析数万条历史摘要，考察女性史在整个领域中占有什么位置、包含哪些主题、在哪些区域史中最常出现、出版数量何时增减以及女性史学者主要关注哪些年代。该论证经过三个尺度：先计算全部摘要中的女性史数量，再用主题模型识别女性史摘要中的主题领域，最后以性史为个案，用图表呈现量化模式。Jo Guldi 也使用主题模型在 Hansard 议会辩论中建立关于基础设施的长期模式，寻找技术、基础设施和建成环境相关词语，以复制、增强或细化先前学者的“枪炮、地图与钢铁”论题。

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Midwestern cities like Chicago in commercial content than any other category. At the regional scale, more of the newspaper focused on Texas and surrounding areas than further afield. Shifting scale to look at where places around Texas appeared, Blevins found them in fragmentary and non-narrative content such as advertisements, which came to prominence due to shifts in American journalism. While the large size of the set of sources used in this argument gives weight to its conclusion, the argument is based on a specific subset of the sources. In that way it differs from other patterns of argument in digital history that examine the relationships of patterns to the larger context, how typical or anomalous they are. Instead the argument looks for patterns within the place names he found and develops by shifting scale to close readings that identify the specific contexts in which those places appear, at the level of sections of the paper rather than within individual stories. Shift in scale fits Blevins' use of word frequency as a method for identifying the concept of space; studies focused on objects more often remain focused on overall patterns.41 Questions about borrowings and relationships between texts are generally framed in terms of specific publications, so they often involve not simply "a question looking for a corpus" but the creation of datasets for analysis. One of the authors, in collaboration with Kellen Funk, created a corpus of 222 statutes consisting of around 180,000 regulations to identify borrowings among those statutes and the extent of the influence of New York's Field Code. Dividing text into individual sections of code and working with five-word groups in each section, they generated similarity scores that identified from which section of a previous code, if any, it was most likely derived, identifying 106,000 borrowings.
The argument of our "The Spine of American Law: Digital Text Analysis and U.S. Legal Practice" followed the structure of shifting scales toward a close reading, with different methods used to identify patterns at different scales of analysis: network analysis, visualizations, and clustering.42 Examining each code as a whole, sect

ion-to-
section borrowings were aggregated to show how many each code borrowed from each other
code, with the results visualized as a network graph that showed the sequence of
borrowing and creation of regional families of codes. Borrowing within each code was
analyzed by visualizing each code as a grid with each box colored to the code from which
it was borrowed. Finally, to analyze individual sections independent of their code, they
were clustered based on similarity to each other and sorted chronologically. That
approach of “taking them out of the context of the codes and putting them into the
context of their particular variations” represented a form of “algorithmic close
reading,” directing attention to variations in sections.43 These analyses identified the
states whose archives should be the focus of research and close reading to explain the
patterns of borrowing. Whereas arguments asking specific questions of a dataset using
word frequency focus on only a subset of the data, this argument about text reuse is
based on similarity scores and patterns within the data as a whole. Analysis of those
patterns develops by progressively narrowing the scale, employing different
visualizations at each scale that could be inspected both in the aggregate and in
detail, and in which change over time is made visible by linking borrowings
sequentially.
In conclusion, we think that the discussion above demonstrates that digital historians
can make meaningful historical interpretations if they chose to do so.
Certainly, despite earlier doubts that they would, digital historians have begun

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Guldi 选择部分主题作为动态主题模型分析，比较不同年代中词语及其相对排序，并用表格阴影突出模式。这些主题模型指向“重要语境”，突出技术、制度联盟和政治结构发生转变的时刻。与词频论证一样，主题模型论证也使用综合数据集理解来源本身性质，而非作为其内部模式的语境。但它们不同于 Hitchcock 和 Turkel 的论证，因为解释通过尺度转换展开：先聚焦特定主题而不是整个文本，再考察随时间的变化；这反映出主题模型最初并不按时间划分，还需要作为动态主题模型进一步分析。计算文本分析的第二条路径则从具体问题进入大数据集，并发展到细读。数字历史学家研究由专有名词标示的对象（人、地点、组织、产品）时，可根据词频构造关于这些对象的论证。Cameron Blevins 测量 Houston Daily Post 页面中特定地名的频率和分布，以考察报纸如何在民族化力量时代建构空间。他的论证基于数字方法才可能获得的证据规模和完整性：1700期报纸及其所有内容的聚合。当地名频率被绘成地图时，报纸想象地理呈现两个尺度：面向纽约和美国中西部的全国尺度，以及以得克萨斯及其邻近区域为主的区域尺度。在全国尺度上，他把“想象”地理置于“实际”地理语境中，突出一些异常，如出现频率超过人口规模预期的城市，以及美国南方的缺席。叠加铁路网络图后可见，报纸中的地点与铁路网络相联系。他再通过改变尺度确认这一点：先细读报纸中反复出现的铁路栏目，再抽样分析每期报纸不同版面内容类型，显示商业内容中纽约和芝加哥等中西部城市相对更常出现。区域尺度上，报纸更多关注得克萨斯及周边地区；细看周边地名出现位置，Blevins 发现它们存在于广告等碎片化、非叙事内容中，而这些内容由于美国新闻业转变而日益重要。

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to make such arguments in their specific fields of study. By drawing together such works
of scholarship for discussion in this essay, we hope to show the possibilities f

or
argumentative scholarship to other digital historians.
But ultimately, digital historians will need to know not just whether such schol arship
is possible but how they should go about it. The articles by Leonardo Barleta an d Rachel
Midura in this special section provide two models for such scholarship. And thei r
annotated articles, along with those of eight other scholars that can found on t he site
Models of Argument-Driven Digital History, illustrate the patterns of argumentat ion
implicit in their historical interpretations. In this introduction, we have atte mpted to
identify successful digital history articles and also to explain the patterns of
argumentation latent in those articles. One core pattern is argument through mor e
comprehensive contextualization: historians put a person or place or text in the context
of geography or network or corpus, or create a context in which to identify typi cal and
anomalous examples and patterns. The transformation of sources into data require d for
digital tools means these contexts can be created out of fragmentary material.
Other patterns of argument flow from creating a broad context of data. One patte rn is
argument through visualization and quantification: historians identify patterns in their
sources by visualizing their data—patterns of spatial proximity in maps, connect ions in
network graphs, and interaction with spaces in 3D models—or by using quantitativ e
measures—calculations of who connected the most individuals in a graph, counts o f word
frequency, topic models of word collocation—or a combination of both. Another pa ttern is
argument through scale:
historians shift scales, exploring patterns in multiple contexts to identify the ir
features and point to material that can be analyzed to interpret their meaning.
A final pattern is argumentation through periodization: digital history that can show
patterns in change over time that naturally connect to the arguments that histor ians
make in periodizing their subjects.
Our point in listing out these patterns is not to create an exhaustive taxonomy of the
ways that digital historians argue. Such a taxonomy would be as limited and rapi dly out
of date for digital history as it would be for conventional history. Rather, we suggest
that thinking in terms of patterns of argumentation is a useful exercise for dig ital
historians. Understanding the structure of arguments that other digital historia ns have
made, including the many examples annotated on the companion website, should hel p
digital historians think through the kinds of arguments that they themselves cou ld make.
We hope that many other digital historians will emulate such patterns and even d evelop
new patterns of argumentation of their own.44 Endnotes The authors wish to thank

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with us and for thinking through the problem of argument and digital history. They
provided the participants with invaluable insight into the publication and peer review
process. Leonardo Barleta, Genevieve Carpio and Andrzej Rutkowski, Mariola Espinosa,
Jacquelyne Howard, Rachel Midura, Erin Sassin and Florence Feiereisen, Lauren Tilton,
and Nathan Tye participated in the workshops

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Blevins 的论证因史料集合巨大而有说服力，但它基于来源中的特定子集；这不同于那些考察模式与更大语境之关系的数字史论证。它在所发现的地名中寻找模式，并通过转向细读来识别这些地点出现的具体语境，尺度是报纸版面栏目而不是单篇故事。文本之间借用和关系问题通常围绕特定出版物提出，因此不仅是“一个问题寻找语料库”，还涉及为分析创建数据集。作者之一与 Kellen Funk
合作建立了包含222部法规、约18万条规定的语料库，以识别法规之间的借用及纽约
Field Code 的影响程度。他们把文本分为法典章节，以五词组为单位生成相似度分数，识别每一节最可能来自哪一部先前法典，最终发现106000处借用。《美国法律的脊梁》一文采用向细读逐步缩小尺度的结构，用网络分析、可视化和聚类在不同尺度识别模式。把每部法典作为整体考察时，章节间借用被聚合为各法典之间的借用数量，并可视化为网络图，显示借用顺序和区域法典家族的形成。每部法典内部借用则以网格可视化，每个方格颜色表示其借自哪部法典。最后，为独立于法典背景分析个别章节，研究者按相似性聚类并按年代排序。这种把章节从法典语境中取出、置于其具体变体语境中的方法，是一种“算法细读”，把注意力引向章节差异。分析识别出哪些州的档案应成为研究和细读重点，以解释借用模式。相比使用词频回答具体问题的论证，这种文本复用论证基于相似度分数和整体数据中的模式；分析通过逐步缩小尺度展开，在每个尺度使用可聚合也可细看的不同可视化，并通过顺序连接借用来显示时间变化。结论认为，数字历史学家确实能够作出有意义的历史解释。尽管早先有人怀疑，数字历史学家已经开始在具体研究领域提出这样的论证。作者汇集这些学术作品，是为了向其他数字历史学家展示论证性学术的可能。

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tioned below. We thank them all for their willingness to take the time to share their
work, and take on the challenge of developing historical arguments. Address
correspondence to Stephen Robertson, George Mason University. Email: srober30@gmu.edu.
Lincoln Mullen, George Mason University. Email: lincoln@lincolnmullen.com.
1. The exception is that, by and large, even conventional historical scholarship has
been transformed by access to databases of primary sources. We are entirely in agreement
with the point of view that searching databases is a form of digital scholarship.
Nevertheless, we still find it worthwhile to draw a distinction between the practices of
conventional historians, however much they have been affected by digital sources and
research techniques, and the practices of digital historians who explicitly identify as
such. See Lara Putnam, “The Transnational and the Text-Searchable: Digitized Sources and

the Shadows They Cast," American Historical Review 121, no. 2 (2016): 377–402, https://doi.
org/10.1093/ahr/121.2.377; Julia Laite, "The Emmet's Inch: Small History in a Digital
Age," Journal of Social History 53, no. 4 (June 1, 2020): 963–89,
https://doi.org/10.1093/ jsh/shy118; and Tim Hitchcock, "Digital Searching and the Re-
formulation of Historical Knowledge," in The Virtual Representation of the Past, ed.
Mark Greengrass and Lorna Hughes (Farnham, 2008), 81–90.
2. Stephen Robertson, "The Differences Between Digital Humanities and Digital History,"
in Debates in Digital Humanities 2016, ed. Matthew Gold and Lauren Klein
(Minneapolis, 2016), https://dhdebates.gc.cuny.edu/read/untitled/section/ed4a1145-7044-
42e9-a898-5ff8691b6628#ch25.
3. Roy Rosenzweig, "Scarcity or Abundance? Preserving the Past in a Digital Age,"
American Historical Review 108, no. 3 (2003): 760.
4. Tom Scheinfeldt, "Where's the Beef? Does Digital Humanities Have to Answer
Questions?" reprinted in Debates in the Digital Humanities, ed. Matthew K. Gold
(Minneapolis, 2012), https://dhdebates.gc.cuny.edu/read/40de72d8-f153-43fa-836ba41d241e9
49c/section/3c03ecdb-2dcf-4597-8fc4-e42f8dcc21e1#p1b2.
5. Ben Schmidt, "Ghost shipping paths" (n.d.): http://benschmidt.org/maps-
visualizations-gallery/.
6. Edward Ayers, Anne Sarah Rubin, William G. Thomas III, and Andrew Torget, The Valley
of the Shadow: Two Communities in the American Civil War,
https://valley.lib.virginia.edu/.
7. Edward Ayers, In the Presence of Mine Enemies: War in the Heart of America, 1859-1863
(New York, 2003); Edward Ayers, The Thin Light of Freedom: The Civil War and
Emancipation in the Heart of America (New York, 2018).
8. Jane Kamensky, The Exchange Artist: A Tale of High-Flying Speculation and America's
First Banking Collapse (New York, 2008).
9. Arguing with Digital History working group, "Digital History and Argument," white
paper, Roy Rosenzweig Center for History and New Media (November 13, 2017), https://
rrchnm.org/argument-white-paper/.
10. See Models of Argument-Driven Digital History, https://model-articles.rrchnm.org/.
11. Robert Darnton, The Great Cat Massacre and Other Episodes in French Cultural History
(New York, 1984); Joan Wallach Scott, "Gender: A Useful Category of Historical
Analysis," The American Historical Review 91 no. 5 (1986): 1053–1075, https://doi.org/10.
2307/1864376.
12. See "Data First" annotation, on Models of Argument-Driven Digital History, https://
model-articles.rrchnm.org/.

最终，数字历史学家不仅需要知道这种学术是可能的，还需要知道如何去做。本特辑中
Leonardo Barleta 和 Rachel Midura
的文章提供了两个模型；它们与另外八位学者带注释的文章一起，在 Models of
Argument-Driven Digital History 网站上展示历史解释中
隐含的论证模式。本文试图识别成功的数字史文章，并解释其中潜在的论证结构。一个核心模式
是通过更全面的语境化来论证：历史学家把人、地点或文本放入地理、网络或语料库语境，或创

造一个能识别典型和异常案例及模式的语境。数字工具要求把来源转化为数据，因此这些语境也可以由碎片化材料建构。其他论证模式来自创造广阔数据语境。其一是通过可视化和量化来论证：历史学家通过把数据可视化来识别来源中的模式，例如地图中的空间邻近、网络图中的连接、3D 模型中的空间互动；或通过量化指标，如谁在网络中连接最多人、词频计数、词语共现主题模型；也可以结合两者。其二是通过尺度来论证：历史学家转换尺度，在多个语境中探索模式，识别特征，并指出可用来解释意义的材料。最后一种是通过分期来论证：能够显示随时间变化模式的数字史，自然能与历史学家围绕研究对象进行分期的论证连接起来。作者列举这些模式并不是要建立穷尽分类；这样的分类对数字史和传统历史学一样都会有限且很快过时。相反，作者认为以论证模式来思考对数字历史学家很有用。理解其他数字历史学家论证的结构，包括配套网站上许多注释案例，应能帮助数字历史学家思考自己可以提出何种论证。尾注部分感谢 Mellon 基金、Journal of Social History 编辑和工作坊参与者，并列出通讯作者与相关参考文献。注释1强调，传统历史学也因可检索数据库而改变，数据库搜索本身也是数字学术，但作者仍区分受数字来源影响的传统历史学家与自认数字历史学家的实践。

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13. Karen Halttunen, “Groundwork: American Studies in Place: Presidential Address to the American Studies Association, November 4, 2005,” American Quarterly 58, no. 1 (2006): 3.
14. Edward Ayers and Scott Nesbit, “Seeing Emancipation: Scale and Freedom in the American South,” Journal of the Civil War Era 1, no. 1 (2011): 3, 10. An annotated version of this article appears on Models of Argument-Driven Digital History, https://modelarticles.rrchnm.org/.
15. Ayers and Nesbit, “Seeing Emancipation,” 14, 17.
16. Scholars involved in the Mapping the Republic of Letters project used map-based visualizations of early modern correspondence networks as the basis for arguments that followed this pattern. Caroline Winterer drew on a map with locations sized for the number of letters sent there in “Where is America in the Republic of Letters?” to “reveal the hidden structures and conditions that nourished the growth of the republic of letters in the early modern period and the causes of its transformation in the nineteenth century,” and provide a new context for that intellectual network and British America’s place in it. See Caroline Winterer, “Where is America in the Republic of Letters?” Modern Intellectual History 9, no. 3 (2012): 597. An annotated version of this article appears on Models of Argument-Driven Digital History, https://model-articles.rrchnm.org/. See also Dan Edelstein and Biliana Kassabova, “How England Fell Off the Map of Voltaire’s Enlightenment article,” Modern Intellectual History 17, no. 1 (2020): 29–53.
17. Harmony Bench and Kate Elswit, “Katherine Dunham’s Global Method and the Embodied Politics of Dance’s Everyday,” Theatre Survey 61, no. 3 (2020): 308. For another example of an argument based on quantifying movement, in this case the voyages of shipping vessels, see Sean Fraga, “Digitally Mapping Commercial Currents: Maritime Mobility, Vessel Technology and U.S. Colonization of Puget Sound, 1851–1861,” Current Rese

arch in
Digital History 3 (2020), https://crdh.rrchnm.org/essays/v03-03-mapping-commerci
al-
currents/.
18. Susannah Ottaway and Auston Mason, “Reconsidering Poor Law Institutions by V
irtually
Reconstructing and Re-Viewing an Eighteenth-Century Workhouse,” The Historical J
ournal
(2020): 10.
19. Ottaway and Mason, “Reconsidering Poor Law Institutions,” 25.
20. That pattern of argument is not limited to projects that use 3D reconstructi
on.
Scholars focused on cities have created more detailed base layers as a context f
or other
data, so the relationship between specific places and events and individuals loc
ated
there can be analyzed. For example, Nicholas Terpstra used DECIMA, a map that vi
sualized
census data from early modern Florence to provide a “socio-economic context” for
 three
sets of data about “where prostitutes lived and worked, what prosecutions they f
aced
most often, and what economic conditions marked their neighbourhoods.” See Nicho
las
Terpstra, “Locating the Sex Trade in the Early Modern City,” in Mapping Space, S
ense and
Movement in Florence, ed. Nicholas Terpstra and Colin Rose (Routledge, 2016), 10
7–
124. See also Catherine Clarke, “Place, identity, and performance: spatial pract
ices and
social proxies in medieval Swansea,” Journal of Medieval History 41, no. 3 (2015
):
256–72.
21. Leo Barleta, “Spatial Genealogies Mobility, Settlement, and Empire-Building
in the
Brazilian Backlands, 1650-1800,” Journal of Social History. An annotated version
 of this
article appears on Models of Argument-Driven Digital History, https://model-arti
cles.
rrchnm.org/.
22. More details of these sources and the process by which Barleta created data
from
them, see the annotation “About Sources” at Models of Argument-Driven Digital Hi
story,
https://model-articles.rrchnm.org/.

注释13至22主要补充空间可视化与地图论证的文献。Karen Halttunen
关于“地点中的美国研究”的演讲提供了空间概念语言。Ayers 和 Nesbit
的《看见解放》是基于空间可视化论证的核心案例。注释16指出，Mapping the
Republic of Letters
项目中的学者也用地图可视化早期现代书信网络，并以此发展论证；Caroline
Winterer 通过按寄信数量调整地点大小的地图，揭示共和国书信网络的隐藏结构及其
在19世纪的转变，同时为英属美洲在该知识网络中的位置提供新语境。注释17列出
Bench 和 Elswit 关于 Katherine Dunham 的研究，并补充
Sean Fraga
对船舶航行和普吉特湾殖民的数字制图研究，说明量化移动也可形成论证。注释18和19列出
Ottaway 与 Mason
关于18世纪济贫院虚拟重建的研究。注释20说明，类似论证模式不限于3D
重建；城市研究者也可以建立细致底图作为其他数据的语境，例如 Nicholas
Terpstra 用 DECIMA 地图把早期现代佛罗伦萨人口普查数据可视化，为妓女

居住和工作地点、遭受起诉类型以及街区经济条件提供社会经济语境。注释21和22指向
Leo Barleta 关于巴西内陆流动、定居与帝国建构的文章及其数据来源说明。

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23. Ruth Ahnert, Sebastian Ahnert, Catherine Nicole Coleman, and Scott Weingart, The
Network Turn: Changing Perspectives in the Humanities (Cambridge, 2020), 7.
24. Robert Morrissey, “Kaskaskia Social Network: Kinship and Assimilation in the French-
Illinois Borderlands, 1695–1735,” William and Mary Quarterly 70, no. 1 (2013): 105.
25. Morrissey, “Kaskaskia Social Network,” 142.
26. Network graphs created from more fragmentary sources produce variations on the
structure of argument employed by Morrissey in addressing the nature of sources as well
as the relationships evident in them. See Maeve Kane, “For Wagrassero’s Wife’s Son:
Colonialism and the Structure of Indigenous Women’s Social Connections, 1690–1730,”
Journal of Early American History 7, no. 2 (2017): 89–114. An annotated version of this
article appears on Models of Argument-Driven Digital History, https://model-articles.
rrchnm.org/. For another example that uses a variety of different sources to identify
the presence of women in Ottoman-Algerian social networks, see Ashley Saunders, “Silent
No More: Women as Significant Political Intermediaries in Ottoman Algeria,” Current
Research in Digital History 3 (2020), https://crdh.rrchnm.org/essays/v03-05-silent-no-
more/
.
27. Ruth Ahnert and Sebastian Ahnert, “Protestant Letter Networks in the Reign of Mary
I: A Quantitative Approach,” ELH 82, no. 1 (2015): 1–33. An annotated version of this
article appears on Models of Argument-Driven Digital History, https://model-articles.
rrchnm.org/.
28. Rachel Midura, “Itinerating Europe: Early Modern Spatial Networks in Printed
Itineraries, 1545–1700,” Journal of Social History–page numbers to come? An annotated
version of this article appears on Models of Argument-Driven Digital History,
https://modelarticles.rrchnm.org/.
29. Midura, “Itinerating Europe,”- page numbers to come?
30. Midura, “Itinerating Europe,”- page numbers to come?
31. Midura, “Itinerating Europe,”- page numbers to come?
32. Midura, “Itinerating Europe,”- page numbers to come?
33. Midura, “Itinerating Europe,”- page numbers to come?
34. For a useful elaboration of the relationship between distant reading (here termed
machine reading) and close reading, see Katherine Hayles, How We Think: Digital Media
and Contemporary Technogenesis (Chicago, 2012), 57–59, 68–75.
35. Tim Hitchcock and William Turkel, “The Old Bailey Proceedings, 1674–1913: Text
Mining for Evidence of Court Behavior,” Law and History Review 34, no. 4 (2016): 929–

55. An annotated version of this article appears on Models of Argument-Driven Digital
History, https://model-articles.rrchnm.org/.
36. Sharon Block and David Newman, “What, Where, When, and Sometimes Why:
Data Mining Two Decades of Women’s History Abstracts,” Journal of Women’s History
23, no. 1 (2011): 82. An annotated version of this article appears on Models of
Argument- Driven Digital History, https://model-articles.rrchnm.org/.
37. Jo Guldi, “Parliament’s Debates about Infrastructure: An Exercise in Using Dynamic
Topic Models to Synthesize Historical Change,” Technology and Culture 60, no. 1
(2019):
3. An annotated version of this article appears on Models of Argument-Driven Digital
History, https://model-articles.rrchnm.org/.
38. Guldi, “Parliament’s Debates,” 9.

中文译文 - 第 17 页

注释23至38主要补充网络、行程表和文本分析案例。Ahnert 等人的 The
Network Turn 提供人文学网络转向的背景。Morrissey 关于
Kaskaskia 社会网络的研究讨论法属伊利诺伊边疆的亲属关系与同化。注释26指出
，更碎片化来源产生的网络图会在 Morrissey 式论证结构上形成变体，并举
Maeve Kane 关于原住民女性社会连接、Ashley Saunders
关于奥斯曼阿尔及利亚女性政治中介的研究为例。注释27列出 Ruth Ahnert 和
Sebastian Ahnert
关于玛丽一世时期新教书信网络的量化研究。注释28至33围绕 Rachel
Midura
关于1545至1700年印刷行程表中的早期现代空间网络的文章，并说明其注释版收录于
Models of Argument-Driven Digital
History。注释34推荐 Katherine Hayles
关于机器阅读和细读关系的讨论。注释35列出 Hitchcock 和 Turkel 对
Old Bailey Proceedings
的文本挖掘研究，用以寻找法庭行为证据。注释36列出 Sharon Block 和
David Newman 对二十年女性史摘要的数据挖掘研究。注释37和38列出 Jo
Guldi 关于议会基础设施辩论的动态主题模型研究。

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1022 39. Guldi, “Parliament’s Debates,” 27.
40. Cameron Blevins, “Space, Nation, and the Triumph of Region: A View of the World from
Houston,” Journal of American History 101 1 (2014): 122–47; and Cameron Blevins,
“Mining and Mapping the Production of Space: A View of the World from Houston,”
(2014), http://web.stanford.edu/group/spatialhistory/cgi-bin/site/pub.php?id¼93.
41. For example, Anne Helmreich, Tim Hitchcock, and William Turkel’s analysis of objects
listed as stolen in trials for theft uses not close reading but two-term phrases
(bigrams) to “mitigate linguistic ambiguity by focusing attention on pairs typically
constituted of an adjective and a noun, such as ’linen handkerchief.” See Anne
Helmreich, Tim Hitchcock, and William Turkel, “Rethinking Inventories in the Digital
Age: The Case of the Old Bailey,” Journal of Art Historiography 11 (2014),
https://arthistoriography.
files.wordpress.com/2014/11/helmreich_hitchcock_turkel.pdf.
42. Kellen Funk and Lincoln Mullen, “The Spine of American Law: Digital Text Analysis
and U.S. Legal Practice,” American Historical Review 123, no. 1 (2018): 132–64.
For another example of an argument about text reuse developed at multiple scales, in

this case in Indian treaties, see Joshua Catalano, “Digitally Analyzing the Uneven
Ground:
Language Borrowing Among Indian Treaties,” Current Research in Digital History 1
(2018), https://crdh.rrchnm.org/essays/v01-02-digitally-analyzing-the-uneven-ground/.
Melodee Beals develops a more iterative approach to shifting scales to explore the
textual reappearance that characterized nineteenth-century newspapers. Her argument
moves from a newspaper corpus to a close reading of five issues then scales the results
of the close reading to reexamine the newspaper as a whole. See Melodee Beals, “Close
Readings of Big Data: Triangulating Patterns of Textual Reappearance and Attribution in
the Caledonian Mercury, 1820–40,” Victorian Periodicals Review 51, no. 4 (2018): 616–39.
An annotated version of this article appears on Models of Argument-Driven Digital
History, https://model-articles.rrchnm.org/.
43. Funk and Mullen, “The Spine of American Law,” 154.
44. See Models of Argument-Driven Digital History, https://model-articles.rrchnm.org/.

注释39至44补充计算文本分析和文本复用案例。Guldi
的研究继续作为主题模型分析基础。Cameron Blevins 关于 Houston
Daily Post 的研究考察地方、国家和区域想象，并有配套的“从休斯顿看世界”的
空间史项目。注释41指出，Anne Helmreich、Tim Hitchcock
和 William Turkel 对 Old Bailey 盗窃审判中被盗物品清单的
分析并非细读，而是使用由形容词和名词组成的二词短语来减少语言歧义。注释42列出
Kellen Funk 和 Lincoln Mullen
的《美国法律的脊梁》，展示数字文本分析与美国法律实践之间的关系；同时补充
Joshua Catalano 关于印第安条约语言借用的多尺度文本复用论证，以及
Melodee Beals
对19世纪报纸文本再出现与署名模式的迭代式尺度转换研究。注释43指出 Funk 和
Mullen 论文中的关键页码，注释44再次引导读者访问 Models of
Argument-Driven Digital History
网站。总体上，尾注展示了本文论证所依赖的跨领域案例网络：地图、3D 重建、空间网络、
社会网络、书信网络、行程表、词频、主题模型和文本复用共同构成数字史论证的可参照范式。